

M.Sc Computer Science Project

Title: Probabilistic Customer Segmentation Using Gaussian Mixture

Model

Abstract

Customer segmentation is an important task in data analytics that helps organizations understand customer purchasing behavior and design targeted marketing strategies. Traditional clustering methods such as k-means assign each customer to a single cluster, which may not effectively represent overlapping or complex customer behavior patterns. To overcome this limitation, this project applies the Gaussian Mixture Model (GMM), a probabilistic clustering technique, for customer segmentation.

In this project, a customer segmentation dataset containing features related to customer spending behavior is used. The system is implemented using the Expectation–Maximization (EM) algorithm, which iteratively estimates the probability of each customer belonging to different clusters and updates the model parameters. The project is developed in a modular manner, including data collection and preprocessing, parameter initialization, expectation step, maximization step, convergence checking, clustering and prediction, visualization, and evaluation.

Data preprocessing techniques such as missing value handling and Z-score normalization are applied to ensure stable model performance. The resulting clusters represent different customer groups based on spending patterns, with each customer having a probability of belonging to multiple segments. Evaluation metrics such as Silhouette Score and BIC/AIC are used to assess clustering quality and to determine the optimal number of customer segments.

The experimental results demonstrate that the Gaussian Mixture Model effectively identifies meaningful and overlapping customer segments compared to hard clustering approaches. This makes GMM suitable for real-world customer segmentation applications, where customer behavior is uncertain and complex.